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Case 27-2025: A 53-Year-Old Man with Embolic Stroke and Left Ventricular Apical Aneurysm

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PRESENTATION OF CASE

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CME



Dr. Jingyi Gong: A 53-year-old man was evaluated in the cardiology clinic of this hospital because of severe left ventricular dysfunction with an apical aneurysm.

Five and a half months before this evaluation, the patient was admitted to the neurology unit of this hospital with multifocal left middle cerebral artery stroke, which had manifested as weakness in the right arm and leg, along with expressive aphasia. These symptoms occurred after diffuse abdominal discomfort that had persisted for 3 days and was associated with heavy consumption of beer; an episode of coffee-ground emesis occurred on the day of admission. Laboratory test results obtained on the day of admission are shown in Table 1; blood levels of calcium, magnesium, phosphorus, albumin, globulin, and bilirubin were normal, as were the results of urinalysis.

Transthoracic echocardiography (Fig. 1A), performed after the administration of an ultrasound-enhancing agent, revealed a dilated left ventricle with an end-diastolic diameter of 64 mm (reference range, 42 to 58) and an ejection fraction of 20% with severe global systolic dysfunction that was most pronounced at the apex. The left ventricular wall thickness was normal. There was evidence of spontaneous echo contrast that was suggestive of a low-flow state at the left ventricular apex, but no discrete thrombus was seen. The left atrium was dilated, and moderate mitral regurgitation was noted, along with aortic sclerosis. There was no evidence of a patent foramen ovale after the administration of agitated saline. Treatment with aspirin, atorvastatin, and apixaban was started.

Dr. Maria Clara Lorca: Inpatient evaluation of cardiac dysfunction included computed tomographic (CT) angiography of the coronary arteries (Fig. 1B, 1C, and 1D), which revealed thinning of the apical myocardium. No evidence of obstructive coronary disease was seen.

Dr. Gong: The results of urine toxicology screening, serum protein electrophoresis, and serum free light-chain testing were normal, as was the thyrotropin level. In addition, the blood ethanol level was undetectable, and screening tests for severe

Table 1. Laboratory Data.*

Variable	Reference Range, Adults†	5.5 Mo before This Evaluation, Hospitalization for Stroke	On Evaluation, Cardiology Clinic Follow-up
Hemoglobin (g/dl)	13.5–17.5	12.3	—
Hematocrit (%)	41.00–53.00	3.89	—
White-cell count (per μ l)	4500–11,000	8100	—
Platelet count (per μ l)	150,000–400,000	280,000	—
Sodium (mmol/liter)	135–145	141	140
Potassium (mmol/liter)	3.4–5.0	4.3	4.2
Chloride (mmol/liter)	98–108	110	101
Carbon dioxide (mmol/liter)	23–32	20	26
Urea nitrogen (mg/dl)	8–25	13	13
Creatinine (mg/dl)	0.60–1.50	0.84	0.83
Glucose (mg/dl)	70–110	135	105
Alanine aminotransferase (U/liter)	10–55	69	27
Aspartate aminotransferase (U/liter)	10–40	68	23
Alkaline phosphatase (U/liter)	45–115	116	73
High-sensitivity troponin T (ng/liter)	0–14	23	—
N-terminal pro-B-type natriuretic peptide (pg/ml)	<900	4406	516
Triglycerides (mg/dl)	0–150	114	—
Low-density lipoprotein cholesterol (mg/dl)	50–129	72	—
High-density lipoprotein cholesterol (mg/dl)	35–100	36	—
Iron (μ g/dl)	30–160	27	—
Total iron-binding capacity (μ g/dl)	230–404	421	—
Ferritin (μ g/liter)	20–300	90	—
Glycated hemoglobin (%)	4.3–5.6	6.4	—
Erythrocyte sedimentation rate (mm/hr)	0–19	15	—
C-reactive protein (mg/liter)	<8.0	3.4	—
D-dimer (ng/ml)	<500	2259	—
Prothrombin time (sec)	11.5–14.5	14.1	14.5
International normalized ratio	0.9–1.2	1.1	1.2
Partial-thromboplastin time (sec)	22.0–36.6	28.6	34.7
Anti- β_2 -glycoprotein 1 IgM (units)	0.0–20.0	<9.4	<9.4
Anti- β_2 -glycoprotein 1 IgG (units)	0.0–20.0	<9.4	<9.4
Anticardiolipin IgM (units)	0.0–15.0	22.3	13.6
Anticardiolipin IgG (units)	0.0–15.0	<9.4	<9.4

* To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for glucose to millimoles per liter, multiply by 0.05551. To convert the values for triglycerides to millimoles per liter, multiply by 0.01129. To convert the values for iron and total iron-binding capacity to micromoles per liter, multiply by 0.1791.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

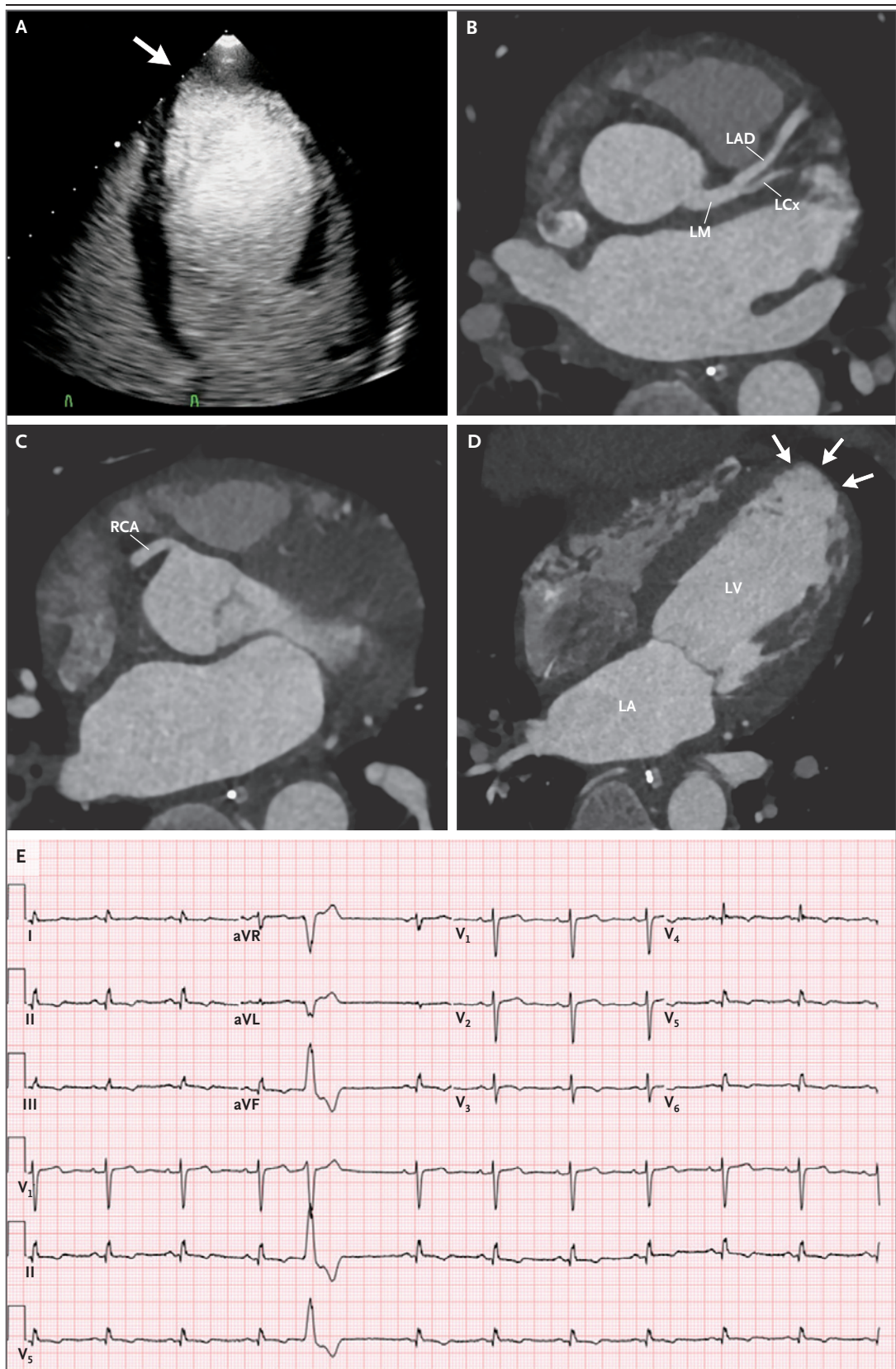


Figure 1 (facing page). Initial Cardiac Studies.

Transthoracic echocardiography and CT coronary angiography were performed at the time of hospitalization for stroke, 5.5 months before the current evaluation. An echocardiogram, obtained after the administration of an ultrasound-enhancing agent (Panel A), shows stasis of blood flow at the left ventricular apex with aneurysmal flow (arrow). Axial images from a CT coronary angiogram (Panels B and C) show an absence of coronary plaques in the left main (LM), left anterior descending (LAD), and left circumflex (LCx) arteries and in the right coronary artery (RCA) (respective distal segments not shown). A CT coronary angiogram obtained in a four-chamber view (Panel D) shows thinning of the apical myocardium (arrows) but no intracardiac thrombus. An electrocardiogram, obtained at the time of the current evaluation (Panel E), shows sinus bradycardia; baseline artifact; borderline-low QRS voltage; T-wave inversions in the inferior, apical, and lateral leads; and a single premature ventricular contraction. These electrocardiogram findings are unchanged from those seen on previous tracings. LA denotes left atrium, and LV left ventricle.

acute respiratory syndrome coronavirus 2, human immunodeficiency virus (HIV) types 1 and 2, hepatitis B virus, and hepatitis C virus were negative.

Given a concern about gastrointestinal bleeding, esophagogastroduodenoscopy was performed, which revealed a clean-based duodenal ulcer with mild erosive gastropathy. *Helicobacter pylori* was detected on examination of the gastric-biopsy specimen and on antigen testing of stool; treatment with omeprazole, bismuth subsalicylate, tetracycline, and metronidazole was initiated. In addition, metoprolol succinate, losartan, spironolactone, and empagliflozin were started for the management of cardiomyopathy. The patient was discharged to an inpatient rehabilitation facility, and he also enrolled in physical therapy and speech therapy for poststroke rehabilitation.

At a follow-up visit in the cardiology clinic 6 weeks after stroke had occurred, the patient reported complete abstinence from alcohol. The dose of beta-blocker was increased, partly owing to short runs of nonsustained ventricular tachycardia noted on an outpatient rhythm monitor. The dose of spironolactone was also increased.

During the next 3.5 months, the patient participated in an outpatient guideline-directed medical therapy (GDMT) program at this hospital. This resulted in adjustments of the doses of metoprolol succinate, spironolactone, and em-

pagliflozin and a change in treatment from losartan to sacubitril–valsartan.

At the current follow-up evaluation in the cardiology clinic, 5.5 months after the initial admission for stroke, the patient reported intermittent lower back pain, the use of two pillows to sleep, and feeling dyspneic if he ate until he felt satiated. He was walking 0.2 km daily. He reported no chest pain, dyspnea at rest, lightheadedness, palpitations, syncope, edema, bleeding symptoms, dysphagia, abdominal pain, focal weakness, or new neurologic symptoms. Before the admission for stroke, the patient had had no known medical conditions and had had no visits for health care in the preceding 10 years. Medications at the current evaluation included apixaban, atorvastatin, metoprolol succinate, spironolactone, sacubitril–valsartan, and empagliflozin. He had no known adverse reactions to medications.

The patient's mother had had hypertension, and his father had had brain cancer. The patient had emigrated from Central America approximately 15 years before this evaluation, and he had worked as a manual laborer before the stroke. He lived with his family in an urban area of New England. For many years before the stroke, he had consumed up to 24 beers per day on weekends, but he had been abstinent from alcohol since the stroke 5.5 months earlier. He had smoked one pack of cigarettes per day for the past 20 years. He had no history of other substance use.

On physical examination, the heart rate was 58 beats per minute, and the blood pressure was 115/75 mm Hg. The weight was 62.6 kg, the height 166.5 cm, and the body-mass index (the weight in kilograms divided by the square of the height in meters) 22.6. Mild facial weakness on the right side and dysarthria were noted. The remainder of the examination was normal. Laboratory test results are shown in Table 1.

An electrocardiogram (ECG) (Fig. 1E) showed sinus bradycardia; baseline artifact; borderline-low QRS voltage; T-wave inversions in the inferior, apical, and lateral leads; intraventricular conduction delay; and a single premature ventricular depolarization. These findings were unchanged from those observed on previous tracings.

Transthoracic echocardiography, performed after the administration of an ultrasound-enhancing agent, showed a left ventricular ejection fraction of 28% with dysfunction that was most pronounced

at the apex, a left ventricular end-diastolic diameter of 52 mm, and mild mitral regurgitation. The left ventricular end-diastolic diameter was smaller than that measured during the patient's initial admission and was now within the normal range. However, other findings were similar to those of the previous echocardiogram.

Dr. Lorca: Given the persistently reduced ejection fraction despite the use of GDMT, a referral for primary prevention with an implantable cardioverter-defibrillator (ICD) was made. Six weeks after the current evaluation, cardiac magnetic resonance imaging (MRI) was performed

before and after the administration of intravenous contrast material (Fig. 2). There was mildly increased left ventricular volume, severely depressed global left ventricular function (ejection fraction, 29%), hypokinesia of the lateral wall with thinning, and dyskinesia of the apex. No hyperintensity was noted on T2-weighted imaging. Extensive late gadolinium enhancement showed circumferential midwall fibrosis; transmural late gadolinium enhancement was noted in the apex, and there was subendocardial late gadolinium enhancement in the lateral territory.

Dr. Gong: A diagnosis was made.

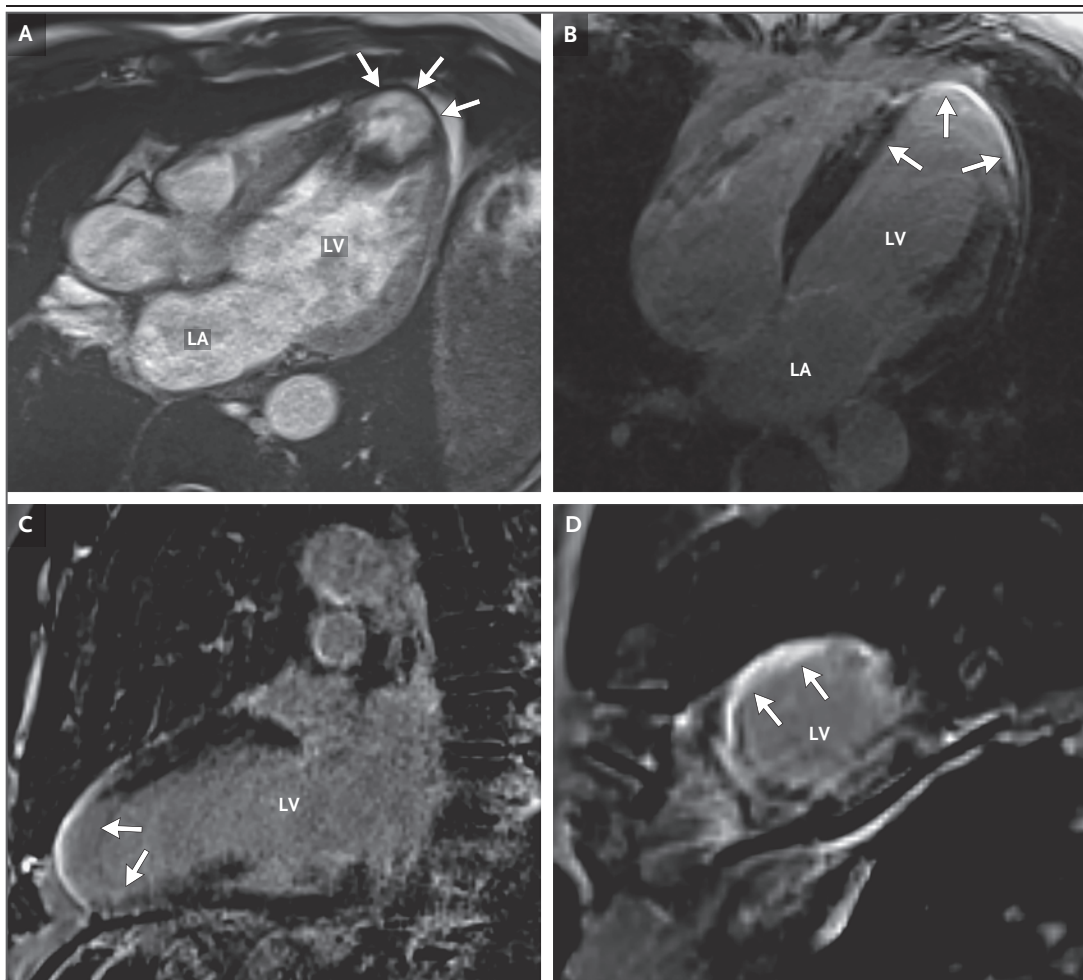


Figure 2. Cardiac MRI.

Cardiac MRI was performed 6 weeks after the current evaluation. A three-chamber balanced steady-state free precession image obtained from cardiac MRI (Panel A) shows thinning of the left ventricular wall (arrows); the presence of dark flow artifact within the left ventricle owing to a low ejection fraction is noted. Late gadolinium enhancement images with phase-sensitive inversion recovery obtained in a four-chamber view (Panel B), a two-chamber view (Panel C), and an apical short-axis view (Panel D) show prominent enhancement (arrows) extending from the midventricle, with subendocardial involvement in the lateral wall and transmural enhancement with wall thinning in the apical region.

DIFFERENTIAL DIAGNOSIS

Dr. Garrick C. Stewart: This 53-year-old man initially presented with embolic stroke and, during evaluation, was found to have dilated cardiomyopathy with a reduced ejection fraction and an aneurysm in the apex of the left ventricle. To establish the underlying diagnosis, we must first explore the nature of the stroke and the causes of cardiomyopathy with aneurysm formation.

CARDIOEMBOLIC STROKE

The patient's presenting stroke was multifocal, which suggests an embolic shower from the heart rather than a vascular origin. Neither ECG monitoring in the hospital nor mobile cardiac telemetry detected atrial fibrillation or flutter, the most common arrhythmic causes of cardioembolic stroke. There was also no evidence of acute myocardial infarction, which can be associated with intracardiac thrombus. Other causes of cardioembolic stroke were ruled out by echocardiography, including right-to-left intracardiac shunting (paradoxical embolus), valvular vegetations, and a cardiac tumor. The patient's enlarged and weak left ventricle with focal dyskinesia at the apex led to stasis of blood flow (part of Virchow's triad of endothelial injury, abnormal blood flow, and hypercoagulability) and thrombus that embolized to the intracranial vessels and therefore was not seen on cardiac imaging.

DILATED CARDIOMYOPATHY

The approach to a patient with newly recognized cardiomyopathy requires careful cardiac evaluation and review of noncardiac coexisting conditions, family history, social history, substance use history, and medication exposures. Whenever a patient is found to have a reduced ejection fraction, coronary artery disease (CAD) must first be ruled out. This is particularly important when established risk factors for atherosclerotic vascular disease are present (e.g., smoking, in this patient's case) because cardiac dysfunction or heart failure may be the initial manifestation of CAD. In this patient, CT angiography effectively ruled out atherosclerosis, along with coronary aneurysm and coronary dissection, as potential causes of ischemia leading to a reduced ejection fraction.

Once CAD is ruled out, the underlying causes of nonischemic dilated cardiomyopathy should be considered in order to inform the prognosis,

select appropriate therapy, and identify at-risk family members. The presence of left ventricular dilatation (as evidenced by the end-diastolic diameter of 64 mm at the time of the initial admission) confirms the presence of chronic — rather than acute — cardiomyopathy, since remodeling (a change in the shape and size of the heart in response to injury or disease) takes months or years to develop. The low voltages and diffuse T-wave inversions that were observed on the initial ECG are nonspecific findings that can occur with many forms of cardiac remodeling, infiltration of the myocardium, or fibrosis. Ventricular dilatation can arise from primary cardiomyopathy, or it can be due to other cardiac causes. Secondary left ventricular dilatation can be caused by chronic tachycardia (long-standing rapid atrial arrhythmias) or electrical dyssynchrony (QRS prolongation or frequent premature ventricular beats), valvular heart disease (especially chronic severe mitral or aortic regurgitation), or adult congenital heart disease. In this patient's case, these possible causes of secondary left ventricular dilatation were ruled out by ECG monitoring and cardiac imaging.

Once primary nonischemic dilated cardiomyopathy is identified in a patient, the broad differential diagnosis includes cardiotoxicity, autoimmunity, stress, nutritional deficiencies, endocrine disorders, inflammatory conditions, infections, causes involving the peripartum period of pregnancy, and idiopathic disorders.¹ In this patient, the modest reduction in the size of the left ventricular chamber after more than 5 months of GDMT is common and does not inform the diagnosis. The patient had a history of heavy alcohol use, which could be a contributing factor, although left ventricular dysfunction persisted after abstinence from alcohol. It is important to note that the patient had no reported history of cocaine use, chemotherapy, or anabolic steroid use and no known exposure to heavy metals.

An autoimmune cause is possible, given the low-level positive anticardiolipin antibody titer, although the patient had no evidence of a connective-tissue disease or other active systemic inflammatory syndrome. However, an inflammatory condition such as remote myocarditis is possible owing to the pattern of fibrosis (late gadolinium enhancement) on cardiac MRI. Cardiac sarcoidosis is unlikely in this patient, given the absence of conduction disease or findings of extracardiac

sarcoidosis, such as hilar lymphadenopathy or interstitial lung disease. Stress cardiomyopathy can manifest as reduced contraction at the cardiac apex relative to the base, but this patient's left ventricular dyskinesia was focal and persisted after many months, which rules out Takotsubo cardiomyopathy. He reported no recent weight loss or "fad" dieting that would suggest nutritional deficiency. He was euthyroid and had no lability in blood pressure that would suggest the presence of a neuroendocrine tumor.

Genetic causes of nonischemic cardiomyopathy must also be considered. These include titin cardiomyopathy due to pathogenic variants in *TTN* (which encodes titin), lamin cardiomyopathy due to variants in *LMNA* (which encodes the lamin A and C proteins), "burnt out" (end-stage) hypertrophic cardiomyopathy (which can be associated with a reduced ejection fraction and thinning of the left ventricular walls), and left ventricular noncompaction. Each of these genetic cardiomyopathies has been linked to increased cardioembolic risk. Even though this patient had no known family history of cardiomyopathy, genetic cardiomyopathies are often underrecognized or can arise from *de novo* mutations.

The patient had no atrial arrhythmias or conduction abnormalities, which typically precede ventricular dysfunction in patients with lamin cardiomyopathy, and the myocardium showed normal compaction on echocardiography and cardiac MRI. Cardiomyopathy associated with muscular dystrophy is unlikely in this patient, given the normal musculoskeletal examination and the absence of iron overload (hemochromatosis) on laboratory testing or cardiac MRI.

Finally, infectious causes must be considered. The patient had a negative test for HIV infection, lived in an area where Lyme disease is endemic (although atrioventricular block was not detected and he had no known tick exposure), and had emigrated from a part of the world (Central America) where common vectors for infection with *Trypanosoma cruzi*, the causative agent of Chagas' disease (also known as American trypanosomiasis), are found.

LEFT VENTRICULAR APICAL ANEURYSM

An important feature of this patient's presentation is the presence of a left ventricular apical aneurysm, since the differential diagnosis narrows

considerably if we focus on overlapping causes of dilated cardiomyopathy and left ventricular apical aneurysm. Cardiac imaging confirmed that he had a true aneurysm with a wide neck and a thinned dyskinetic wall, rather than a pseudoaneurysm (a contained rupture after acute myocardial infarction) or a diverticulum (a focal outpouching that still thickens normally during contraction). Myocardial infarction involving the left-anterior-descending artery distribution, which usually feeds the apical myocardium, is the most common cause of apical aneurysm with associated ventricular dysfunction; however, this diagnosis was ruled out. The patient had no evidence of hypertrophic cardiomyopathy (wall thickness ≥ 15 mm, or ≥ 13 mm with family history), a condition in which apical aneurysms can occur in traditional sarcomeric forms or their phenocopies (especially in patients with Fabry's disease).

Ventricular aneurysms are common in patients with Chagas' cardiomyopathy, and most such aneurysms occur in the left ventricular apex.² This patient had emigrated 15 years ago from Central America, where Chagas' disease is prevalent. Moreover, acute infection with *T. cruzi* can precede the chronic cardiomyopathic phase by years or decades. Later-stage Chagas' cardiomyopathy is often manifested by a dilated ventricle, an apical aneurysm, thromboembolism, and electrical disturbance; this patient had three of these classic findings.³ The absence of right bundle-branch or fascicular block does not rule out Chagas' cardiomyopathy, and indeed, T-wave abnormalities are common with this disease.⁴ The ECG changes noted in Figure 1E are meaningful, albeit subtle, because they are consistent with the presence of the apical aneurysm (mild ST-segment elevation in the inferior, apical, and lateral leads) and are consistent with Chagas' cardiomyopathy (intraventricular conduction delay, which is a potential prelude to a more typical presentation of right bundle-branch block).

Chagas' disease can be manifested by other systemic findings in the nervous and gastrointestinal systems, although this patient's gastrointestinal abnormalities were related to confirmed peptic ulcer disease due to *H. pylori* infection. Genetic dilated or inflammatory cardiomyopathies rarely lead to apical aneurysm formation, and both these possibilities would have been higher on the differential diagnosis had this pa-

tient not emigrated from a region where Chagas' disease is endemic.

Therefore, the diagnosis that best explains the patient's nonischemic dilated cardiomyopathy and left ventricular apical aneurysm is chronic infection with *T. cruzi*. To establish this diagnosis, I would recommend serologic testing to detect the presence of *T. cruzi*-specific antibodies.

CLINICAL IMPRESSION

Dr. Gong: When we evaluated this patient, we thought that the most likely diagnoses that would explain the nonischemic dilated cardiomyopathy with left ventricular apical aneurysm were alcohol use, familial cardiomyopathy, and Chagas' cardiomyopathy, especially given his epidemiologic risk. We also considered the possibility of emboli that had traveled to the left-anterior-descending artery from a presumed apical thrombus, but the normal levels of cardiac biomarkers and the absence of active ischemic symptoms at the time of stroke made this diagnosis unlikely. The patient was referred for genetic analysis, which tested for 82 genes associated with dilated cardiomyopathy and left ventricular noncompaction. Serologic testing for *T. cruzi* antibodies was also performed.

CLINICAL DIAGNOSIS

Chagas' disease.

DR. GARRICK C. STEWART'S DIAGNOSIS

Chronic Chagas' cardiomyopathy.

DIAGNOSTIC TESTING

Dr. Briana Castillo: Genetic testing did not detect any pathogenic or likely pathogenic variants associated with dilated cardiomyopathy. However, the diagnosis in this patient's case was established when the serologic testing for *T. cruzi* antibodies showed a positive result.

T. cruzi is a single-cell protozoan parasite that cycles between the trypomastigote and amastigote forms of the parasite in the mammalian host. The diagnosis of Chagas' disease can be made by direct visualization of the parasite in the peripheral blood during the acute phase of

the disease or by detection of antibodies to the parasite in the chronic phase of the disease.⁵

During the acute phase of Chagas' disease, trypomastigotes can be found circulating in the bloodstream for approximately 8 weeks. Parasites can be directly visualized on microscopic examination of a peripheral-blood or buffy-coat smear with Giemsa staining. Alternatively, polymerase-chain-reaction (PCR) testing of whole blood or cerebrospinal fluid for *T. cruzi* DNA can be used for diagnosis. Given its increased sensitivity over direct microscopy, a PCR test can be used to detect reactivation in immunocompromised patients with chronic Chagas' disease or to monitor for acute *T. cruzi* infection in the recipient of an infected transplanted organ.⁶

During the chronic phase (the indeterminate form) of Chagas' disease, the parasite is no longer circulating in the bloodstream, and diagnosis relies on the detection of IgG to *T. cruzi*. The parasite shows high antigenic diversity among strains, and the sensitivity and specificity of each serologic test varies across populations and regions where Chagas' disease is endemic.⁵ No single assay is considered to have sufficiently high sensitivity and specificity to be relied on alone.⁶ Because of this phenomenon, the World Health Organization (WHO) guidelines recommend two distinct positive serologic tests with different antigens and methods for the diagnosis of chronic Chagas' disease.⁷ In the United States, only four serologic assays have been approved by the Food and Drug Administration for diagnostic use.⁸ If the results between two assays are discordant, a third "tiebreaker" assay is required.

At our hospital, we partner with a reference laboratory that uses the Wiener Laboratories enzyme-linked immunosorbent assay.⁸ If the antibody is detected on the first test, a Chagas' lateral flow assay (the InBios Chagas Detect Plus Lateral Flow Assay) is performed reflexively. In this patient, both assays were positive, which satisfied the WHO requirement for diagnosis of chronic Chagas' disease with two serologic tests that involved different antigen profiles and methods.^{7,8}

LABORATORY DIAGNOSIS

Chronic infection with *Trypanosoma cruzi*.

DISCUSSION OF MANAGEMENT

Dr. Barbra M. Blair: Treatment of Chagas' disease is determined by the stage of infection and should be offered to patients with acute or congenital disease, to patients with reactivation of disease, and to children whose disease is in the chronic phase (the indeterminate form). In addition, given that the detection and treatment of Chagas' disease early during the indeterminate period can prevent late complications, adults 50 years of age or younger without cardiac involvement should be offered treatment. For patients older than 50 years of age, most experts agree that, despite insufficient data, treatment can be considered on a case-by-case basis. For patients meeting the criteria to receive treatment, either benznidazole or nifurtimox is administered for 60 days.

Given that this patient was older than 50 years of age, there is no compelling evidence to suggest that he would benefit from antiparasitic therapy. The patient should be counseled to avoid donating blood and to have younger family members who lived in the same rural setting screened for infection with *T. cruzi*, especially any family members who can become pregnant. Ongoing care of this patient should involve management of his cardiomyopathy and, if indicated, evaluation for cardiac transplantation. If the patient were to undergo transplantation, then treatment and monitoring for reaction to infection would be indicated.

Dr. Erin E. Coglianese: The management of the patient's heart failure and cardiomyopathy center around GDMT, prophylaxis against sudden death from cardiac causes, consideration of medical devices to optimize the management of heart failure, and risk stratification to determine whether advanced heart failure therapies are needed.⁹ GDMT is the cornerstone of management of cardiomyopathy. This therapy includes the use of four medications: a beta-blocker (carvedilol, metoprolol succinate, or bisoprolol); an angiotensin receptor–neprilysin inhibitor, or an angiotensin-receptor blocker or an angiotensin-converting-enzyme inhibitor (if there is a contraindication to the angiotensin receptor–neprilysin inhibitor); a mineralocorticoid receptor antagonist; and a sodium–glucose cotransporter 2 inhibitor. Other agents that could be considered include ivabradine (which is used for patients who have a persistently elevated heart

rate that is not attributed to low cardiac output), soluble guanylate cyclase stimulators, and vasodilators such as hydralazine and isosorbide dinitrate. The combination of GDMT agents has been shown to reduce the number of episodes of worsening heart failure and improve survival.

In addition to medication management, patients should receive education regarding the monitoring of their volume status (daily measurement of body weight) and the use of diuretic agents to maintain an appropriate volume status. Recommendations regarding dietary sodium intake and fluid restrictions should be individualized for each patient.

ICDs have resulted in an improvement in survival among patients with ischemic and nonischemic cardiomyopathy by preventing death from ventricular tachycardia and ventricular fibrillation, as shown in the MADIT-II (Multicenter Automatic Defibrillator Implantation Trial II) and SCD-HeFT (Sudden Cardiac Death in Heart Failure Trial).^{10,11} Patients who have been receiving GDMT for 3 months, or for more than 90 days after revascularization with a left ventricular ejection fraction of 35% or less, are candidates for ICDs.

In addition to ICDs, several other therapies that may reduce symptoms and improve the left ventricular ejection fraction in patients with cardiomyopathy can be considered. These include cardiac resynchronization therapy in patients with left bundle-branch block, cardiac contractility modulation, and baroreflex activation therapy.¹² Although the goal of GDMT and devices to treat heart failure is to stabilize symptoms, some patients ultimately progress to end-stage heart failure and undergo evaluation for advanced heart-failure therapies including cardiac transplantation. For this patient, careful dose adjustment of GDMT, appropriate selection of medical devices, and early detection of the transition to advanced heart failure (should this occur) may improve his long-term survival.

FOLLOW-UP

Dr. Gong: The patient continues to have follow-up visits at the cardiology clinic for management of chronic heart failure. Echocardiography performed 1 year after his initial admission showed recovery of left ventricular systolic function (ejection fraction, 42%), with a left ventricular end-

diastolic diameter of 50 mm. The left ventricular apical aneurysm was still present, so the patient's anticoagulant therapy was continued for secondary prevention of recurrent stroke. Mobile cardiac telemetry showed three episodes of nonsustained ventricular tachycardia (up to seven consecutive beats); given these findings, along with the known elevated risk of sudden death from cardiac causes in the context of Chagas' cardiomyopathy, he was referred to the electrophysiology laboratory for consideration of electrophysiologic testing and possible placement of an ICD.¹³

FINAL DIAGNOSIS

Chagas' disease.

This case was presented at the Medicine Case Conference.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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REFERENCES

1. Bozkurt B, Colvin M, Cook J, et al. Current diagnostic and treatment strategies for specific dilated cardiomyopathies: a scientific statement from the American Heart Association. *Circulation* 2016;134(23):e579-e646.
2. Oliveira JS, Mello De Oliveira JA, Frederigue U Jr, Lima Filho EC. Apical aneurysm of Chagas's heart disease. *Br Heart J* 1981;46:432-7.
3. Nunes MCP, Beaton A, Acquatella H, et al. Chagas cardiomyopathy: an update of current clinical knowledge and management: a scientific statement from the American Heart Association. *Circulation* 2018;138(12):e169-e209.
4. Ribeiro AL, Sabino EC, Marcolino MS, et al. Electrocardiographic abnormalities in *Trypanosoma cruzi* seropositive and seronegative former blood donors. *PLoS Negl Trop Dis* 2013;7(2):e2078.
5. Pérez-Molina JA, Molina I. Chagas disease. *Lancet* 2018;391:82-94.
6. Bern C, Kjos S, Yabsley MJ, Montgomery SP. *Trypanosoma cruzi* and Chagas' disease in the United States. *Clin Microbiol Rev* 2011;24:655-81.
7. Pan American Health Organization. Guidelines for the diagnosis and treatment of Chagas disease. 2019 (<https://www.paho.org/en/documents/guidelines-diagnosis-and-treatment-chagas-disease>).
8. Kelly EA, Bulman CA, Gunderson EL, et al. Comparative performance of latest-generation and FDA-cleared serology tests for the diagnosis of Chagas disease. *J Clin Microbiol* 2021;59(6):e00158-21.
9. Maddox TM, Januzzi JL Jr, Allen LA, et al. 2024 ACC expert consensus decision pathway for treatment of heart failure with reduced ejection fraction: a report of the American College of Cardiology Solution Set Oversight Committee. *J Am Coll Cardiol* 2024;83:1444-88.
10. Moss AJ, Zareba W, Hall WJ, et al. Prophylactic implantation of a defibrillator in patients with myocardial infarction and reduced ejection fraction. *N Engl J Med* 2002;346:877-83.
11. Bardy GH, Lee KL, Mark DB, et al. Amiodarone or an implantable cardioverter-defibrillator for congestive heart failure. *N Engl J Med* 2005;352:225-37.
12. Sharif ZI, Galand V, Hucker WJ, Singh JP. Evolving cardiac electrical therapies for advanced heart failure patients. *Circ Arrhythm Electrophysiol* 2021;14(4):e009668.
13. Keegan R, Yeung C, Baranchuk A. Sudden cardiac death risk stratification and prevention in Chagas disease: a non-systematic review of the literature. *Arrhythm Electrophysiol Rev* 2020;9:175-81.

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